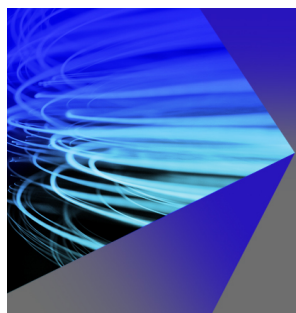


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ABSTRACT

Two-dimensional (2D) carbon allotropes for high-performance electrode materials in lithium-ion batteries (LIBs) are attracting a great deal of research interest due to their unique architectures. Herein, we perform the first-principles calculations to systematically investigate the feasibility of 2D sp-sp² hybridized THD-C (tetra-, hexa-, and dodeca-membered rings), as a potential anode for LIBs. Our simulations indicate that two spatially adjacent acetylenic bonds engender a modest Li-C coupling via the cooperative cation- π interaction driven by partial electron transfer, which exhibits a suitable Li adsorption energy of -0.78 eV. 2D THD-C is proven to be a favorable anode material that offers a high theoretical capacity (1116.7 mAh/g), a low Li diffusion barrier (0.49 eV), and a moderate open-circuit voltage (0.40 V). Furthermore, it retains a small volume change (~3%) as well as favorable structural and thermal stability even at maximum Li concentration. Overall, those findings not only provide an in-depth mechanistic insight but also extend an effective method to search for high-performance anode materials in advancing LIB technology.

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Lithium-ion batteries (LIBs) have become the dominant energy storage technology for portable electronics and electric vehicles owing to their high reversibility, suitable working voltage, and excellent reliability.¹⁻⁵ The electrochemical performance of LIBs largely depends on the properties of the anode materials. At present, graphite remains as a widely used anode due to its high stability and low cost; however, its low theoretical capacity (372 mAh/g) severely limits further improvements in energy density.⁶ Therefore, developing high-capacity anodes is of great significance for next-generation LIBs.

Various two-dimensional (2D) carbon allotropes have been theoretically predicted or experimentally synthesized, such as QH-graphene,⁷ Ψ -graphene,⁸ biphenylene,⁹ FSL-graphene,¹⁰ tolanene,¹¹ and graphdiyne (GDY).¹² These ring-diverse networks, assembled from nonhexagonal carbon rings and composed of sp² or mixed sp-sp² hybridized frameworks, exhibit unique electronic delocalization and adjustable pore geometries that enhance ion adsorption and migration.^{7,13-15} The introduction of sp-hybridized carbon atoms via acetylenic linkages into conjugated sp² networks can significantly modulate the charge distribution and enhance the Li-C coupling strength. In particular, the unique coexistence of sp-sp² hybridized carbon atoms in GDY endows the electrode with tunable electronic characteristics, strengthening the coupling effect

and thereby enhancing the theoretical capacity and optimizing the electrochemical performance.¹⁶⁻¹⁹ Such mixed-hybridization systems not only provide multiple active adsorption sites but also facilitate fast charge transfer, making them promising candidates for high-performance anode materials beyond graphene-based systems.

Recently, Jenkins *et al.* proposed a two-dimensional carbon allotrope with mixed sp-sp² hybridization, termed THD-C (tetra-, hexa-, and dodeca-membered rings), which consists of 12-, 6-, and 4-membered carbon rings.²⁰ The reported synthesis scheme of 2D THD-C is considered to be potentially accessible from halogenated 4-ethynylphenylacetylene molecular precursors via on-surface Ullmann coupling followed by interpolymer dehydrofluorination (HF-zipping). Previous studies have confirmed its thermodynamic stability and suggested its potential for gas adsorption and metal-ion storage. However, its lithium storage capability and the underlying mechanism have not yet been systematically investigated.

In this work, we evaluate the potential of THD-C as an anode material for LIBs through first-principles calculations. THD-C exhibits competitive lithium storage performance, featuring a suitable adsorption energy, low diffusion barrier, and moderate open-circuit voltage with favorable thermal stability at high Li concentration. Notably, its

theoretical capacity reaches 1116.7 mAh/g, nearly 3 times that of commercial graphite, highlighting its promise for high-capacity batteries. By analyzing its electronic structure evolution before and after lithiation, we reveal the Li storage mechanism within the $sp-sp^2$ hybridized carbon framework. These findings demonstrate the high potential of THD-C for lithium storage and provide a mechanistic basis for designing next-generation carbon-based electrodes with tunable electronic properties.

All density functional theory (DFT) calculations were performed in the Vienna *ab initio* simulation package (VASP) using the projected augmented wave (PAW) method with a plane wave basis.^{21–23} The exchange–correlation effects were described by the Perdew–Burke–Ernzerhof functional (PBE) within the generalized gradient approximation (GGA).^{24,25} van der Waals (vdW) interactions were included using DFT-D3.²⁶ A vacuum layer of 30 Å was applied along the z-direction to eliminate spurious interlayer interactions. The Brillouin zone was sampled using a $13 \times 7 \times 1$ Monkhorst–Pack k-point grid.²⁷ A plane wave cutoff energy of 600 eV was set, with convergence criteria for the Hellmann–Feynman force and energy at 0.01 eV/Å and 1×10^{-5} eV, respectively. Bader charges were performed to evaluate charge transfer.^{28,29} The crystal orbital Hamilton populations (COHP) analysis was conducted using the LOBSTER package.³⁰ *Ab initio* molecular dynamics (AIMD) simulations verified the kinetic stability of pristine and lithiated THD-C in the NVT ensemble with the temperature controlled by a Nosé–Hoover thermostat.³¹ The dynamic stability was confirmed via density functional perturbation theory (DFPT) implemented in the Phonopy code.^{32,33} The diffusion barriers of Li ions were obtained using the climbing image nudged elastic band (CI-NEB) method.³⁴

As illustrated in Fig. 1(a), the optimized structure of THD-C is a tetragonal lattice in Pmm2 symmetry with lattice constants $a = 3.76$ Å, $b = 7.11$ Å, and $\alpha = \beta = 90^\circ$, which are in good agreement with a previous report.²⁰ The sp -hybridized acetylenic bond (B1) between

adjacent twelve-membered rings is 1.22 Å, shorter than other sp^2 hybridization bonds. In order to quantify the stability of the THD-C configuration, AIMD simulations and phonon dispersion are performed to evaluate its thermal and dynamical stability, respectively. Figure 1(b) reveals that THD-C remains structurally intact without fracture and abnormal energy fluctuations at 1000 K, supporting its thermal stability. Further phonon calculations demonstrate that the THD-C is dynamically stable due to the absence of imaginary frequencies, consistent with a previous report [Fig. 1(c)].²⁰

The electronic property of anode materials directly influences their electrochemical performance in LIBs. We perform atomic orbital projected band structures and projected density of states (PDOS). As presented in Fig. 2(a) and Fig. S2, the metallicity of THD-C mainly arises from the carbon p_z orbitals contributing to the bands near the Fermi level, presenting analogous characteristics to the π states of graphene, while the σ orbitals are distributed at deeper energy levels.^{35,36} Figure 2(b) illustrates that the three nonequivalent carbon atoms in THD-C exhibit distinct p_z orbital contributions near the Fermi level. This metallicity, dominated by the p_z orbitals, facilitates lithium-ion adsorption and charge transfer.¹¹ The electron localization function (ELF) reveals that nearest-neighbor C atoms form covalent bonds and shows significant electron localization at the acetylenic bond positions [Fig. 2(c)]. Band projection analysis indicates the presence of an inclined Dirac cone slightly above the Fermi level near the Y-point, with the corresponding Fermi velocity in the range of 0.835–0.854 $\times 10^6$ m/s, comparable to graphene.^{37–39} This result implies that THD-C possesses good electronic transport capability.

To systematically investigate the lithiation behavior, a $3 \times 2 \times 1$ supercell of THD-C is constructed to verify the adsorption behavior of a single Li atom. Eleven possible Li adsorption sites are considered, as shown in Fig. 1(a). The adsorption energy is calculated by

$$E_{ad} = E_{Li+THD-C} - E_{THD-C} - E_{Li}, \quad (1)$$

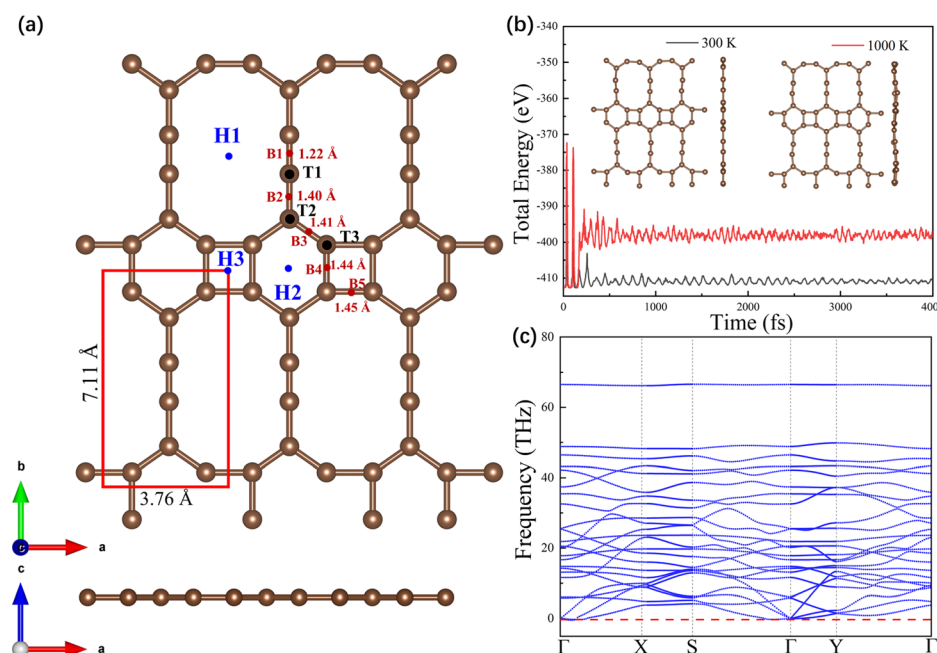


FIG. 1. (a) Top and side views of THD-C. The red line exhibits the unit cell. (b) Total energy fluctuation curve of THD-C at 300 and 1000 K during AIMD simulations. These insets show the snapshots for the structures. (c) Phonon dispersion curves of THD-C.

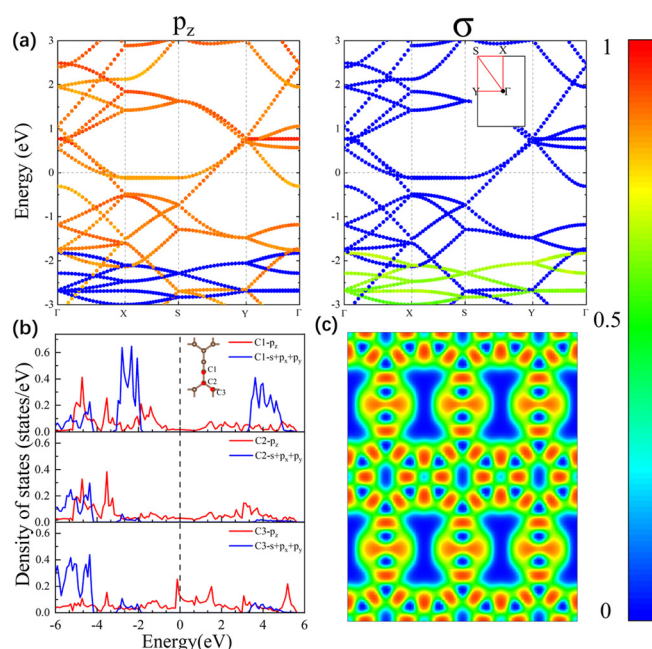


FIG. 2. (a) Electronic band structure of THD-C. The bands are colored by the k-resolved PDOS summed over all p_z (left) and σ (right) orbitals. The Fermi level is aligned to be 0 eV. (b) PDOS of inequivalent C atoms. (c) ELF map of THD-C.

where $E_{\text{THD-C}}$ and $E_{\text{Li+THD-C}}$ are the energy of pristine and Li-adsorbed THD-C, respectively. E_{Li} is the energy of a Li atom in bulk lithium metal (-1.89 eV in this work).⁴⁰ When $E_{\text{Li+THD-C}}$ is negative, the adsorption process is exothermic and thus occurs

spontaneously.^{41,42} Geometry optimization demonstrates that only four adsorption sites remain stable, namely H1, H2, H3, and T1. As illustrated in Fig. 3, the H1 site exhibits the most negative adsorption energy (-0.78 eV) and the shortest vertical distance (1.06 Å). The effective ring size follows the order $\text{H1} (5.825 \text{ Å}) > \text{H2} (3.082 \text{ Å}) > \text{H3} (2.058 \text{ Å})$, consistent with the trend of adsorption energies, suggesting that larger cavities favor stronger Li binding.⁴³ Interestingly, the geometrical preference of larger rings implies that electronic factors follow a similar trend.

We further analyze the p_z orbital occupancy of carbon atoms to elucidate the electronic origin of the adsorption energy differences (Table S1). Due to the introduction of sp-hybridized acetylenic carbon atoms, THD-C manifests an asymmetric distribution of p_z electrons. In contrast to graphene with nearly uniform p_z orbitals, THD-C possesses stronger Li adsorption, suggesting that such p_z nonuniformity strengthens Li binding.⁴⁴ Figure 3 presents that the average p_z occupations in different ring types follow the order $\text{H1} (0.49 \text{ e}) > \text{H2} (0.47 \text{ e}) > \text{H3} (0.45 \text{ e})$, which shows a clear positive correlation with the adsorption energy. This implies that a higher average p_z electron occupancy corresponds to stronger Li adsorption.

Meanwhile, charge density difference maps between Li and THD-C are plotted to gain insight into the nature of the interaction qualitatively. At the H1 site, the spatial distribution of electron density exhibits laterally symmetric overlap with noticeable electron accumulation along both sides of the two spatially adjacent acetylenic bonds, indicating charge transfer between Li and the acetylenic linkages (Fig. 3). Such an interaction between the cation and the acetylenic π system bears a resemblance to cation- π interaction, which has also been observed in previous studies.^{12,45} Bader charge analysis reveals a total charge transfer of $0.910 |e|$ at the H1 site, with an equal amount of charge distributed toward the carbon atoms of the two adjacent

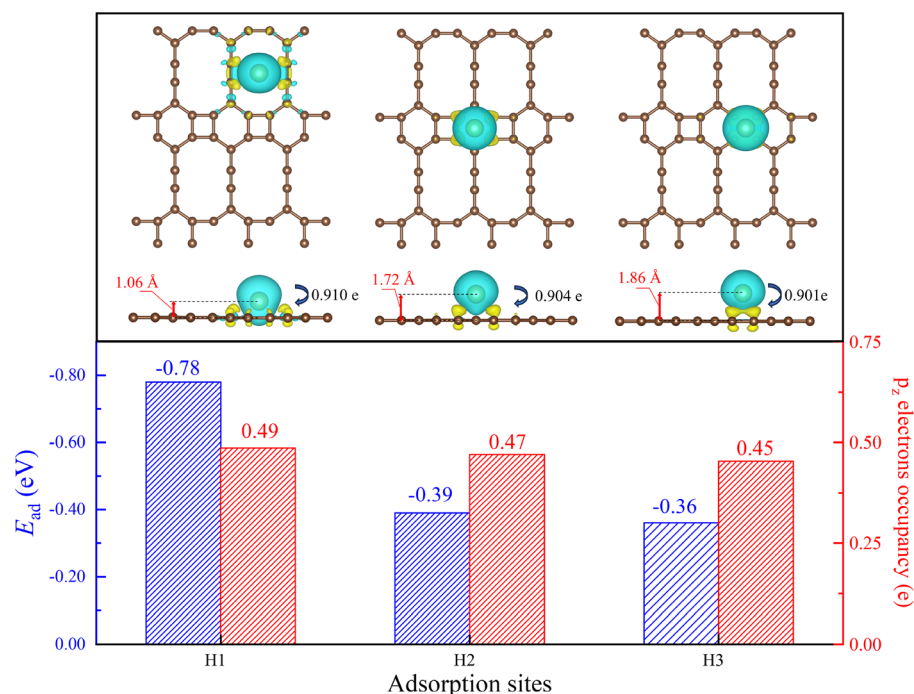


FIG. 3. Top and side views of charge density difference for Li adsorbed at H1, H2, and H3 sites with their adsorption height, respectively. The yellow and blue regions denote charge depletion and accumulation, respectively, with an iso-value of 0.0032 e Å^{-3} . The bar chart shows adsorption energy (E_{ad}) of three adsorption sites and p_z (π) electrons of different ring types of THD-C.

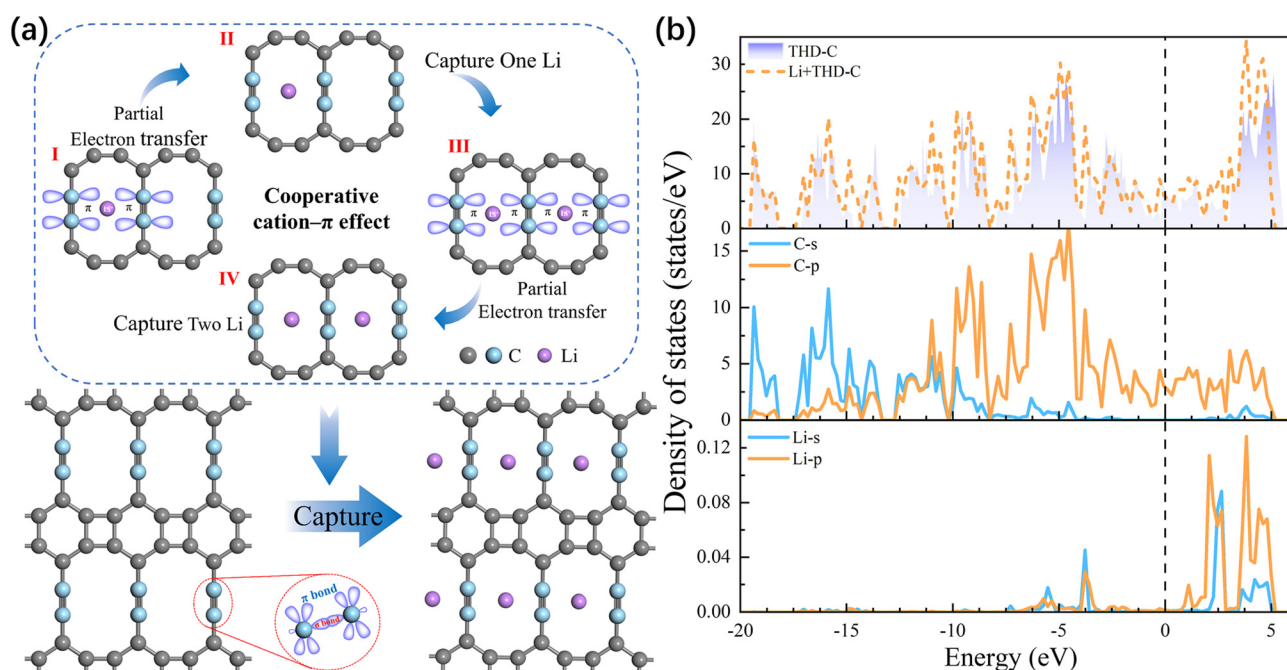


FIG. 4. (a) Schematic diagram of the Li capture process in the Li storage mechanism of THD-C. (b) TDOS of pristine and Li-adsorbed THD-C, respectively, as well as the PDOS of C and Li. The Fermi level is aligned to be 0 eV.

acetylenic bonds. This behavior indicates that THD-C captures Li ions through Li-C coupling formed by partial electron transfer within two spatially adjacent acetylenic π -electron clouds, unlike the single-bond cation- π interaction observed in GDY [Fig. 4(a)].¹² The cation- π interaction energy and charge transfer of adsorbed Li are calculated at the single acetylenic bond (T1 site) to evaluate its interaction strength (Fig. S3). The results show that the cation- π interaction at the H1 site is nearly 3 times that at the single-bond site. Such a finding indicates that the cooperative cation- π interaction arising from adjacent acetylenic bonds enhances the adsorption capability toward Li ions.

To further elucidate the orbital origin of the cooperative cation- π interaction, the total and projected density of states before and after lithiation are calculated. As shown in Fig. 4(b), a weak hybridization occurs between C-p and Li-s orbitals across -2.8 to -6.3 eV. Meanwhile, significant hybridization with C orbitals occurs above the Fermi level (2.1 – 5.2 eV). The COHP and integrated COHP (ICOHP) analyses are further performed to directly evaluate the bonding characteristics of Li-C interactions. Figure S4 shows that weak bonding contributions exist below the Fermi level and pronounced antibonding features exist near and above it. The ICOHP (-0.267 eV) denotes a weak noncovalent interaction compared with typical metal carbon bonds.⁴⁶ The COHP results show that the Li-C interactions are dominated by ionic components with limited orbital overlap and thus are essentially characteristic of cation- π noncovalent interaction.⁴⁷ These findings indicate that the π system of sp-hybridized carbon forms a cooperative cation- π interaction through partial charge transfer with Li ions.

Li-ion diffusion dictates the rate capability of LIBs. As shown in Figs. 5(a)–5(c), three equivalent diffusion pathways exist for Li migration between adjacent H1 sites. The CI-NEB calculations yield energy

barriers of 0.49, 0.64, and 0.58 eV for paths I, II, and III, respectively [Figs. 5(b)–5(d)]. Notably, path I possesses the lowest diffusion energy barrier, which is comparable to those of Me-graphene (0.48 eV),⁴⁸ graphdiyne (0.52 eV),⁴⁹ and tolanene (0.61 eV).¹¹ These results suggest that THD-C exhibits moderate and favorable Li diffusion kinetics among carbon-based materials. The diffusion coefficient (D) is further estimated using the Arrhenius relation as follows:⁵⁰

$$D \sim \exp\left(-\frac{E_a}{k_b T}\right), \quad (2)$$

where E_a represents the diffusion barrier energy. $k_b T$ is taken as 0.026 eV.⁴² The diffusion coefficients along paths I, II, and III are calculated to be 9.59×10^{-9} , 2.04×10^{-11} , and 2.05×10^{-10} , respectively. The results demonstrate that the diffusion coefficient along path I is approximately 470 times that of path II and 47 times that of path III, which means Li preferentially diffuses along path I.

In evaluating the theoretical capability, single-layer adsorption is examined by successively introducing Li atoms into the available sites of a $3 \times 2 \times 1$ supercell. When the Li-Li distance falls below ~ 2.67 Å, dimers form and reduce storage efficiency.⁴¹ To prevent clustering, H2 sites are occupied after fully filling the H1 sites, while the H3 sites remain unoccupied, consistent with lithiation behavior reported for biphenyl-like carbon rings.^{11,43} The average adsorption energy formula used here is

$$E_{ave} = \frac{(E_{nLi+THD-C} - E_{THD-C} - nE_{Li})}{n}, \quad (3)$$

where $E_{nLi+THD-C}$ is the total energy of the supercell with n adsorbed Li atoms, E_{THD-C} is the energy of the pristine substrate,

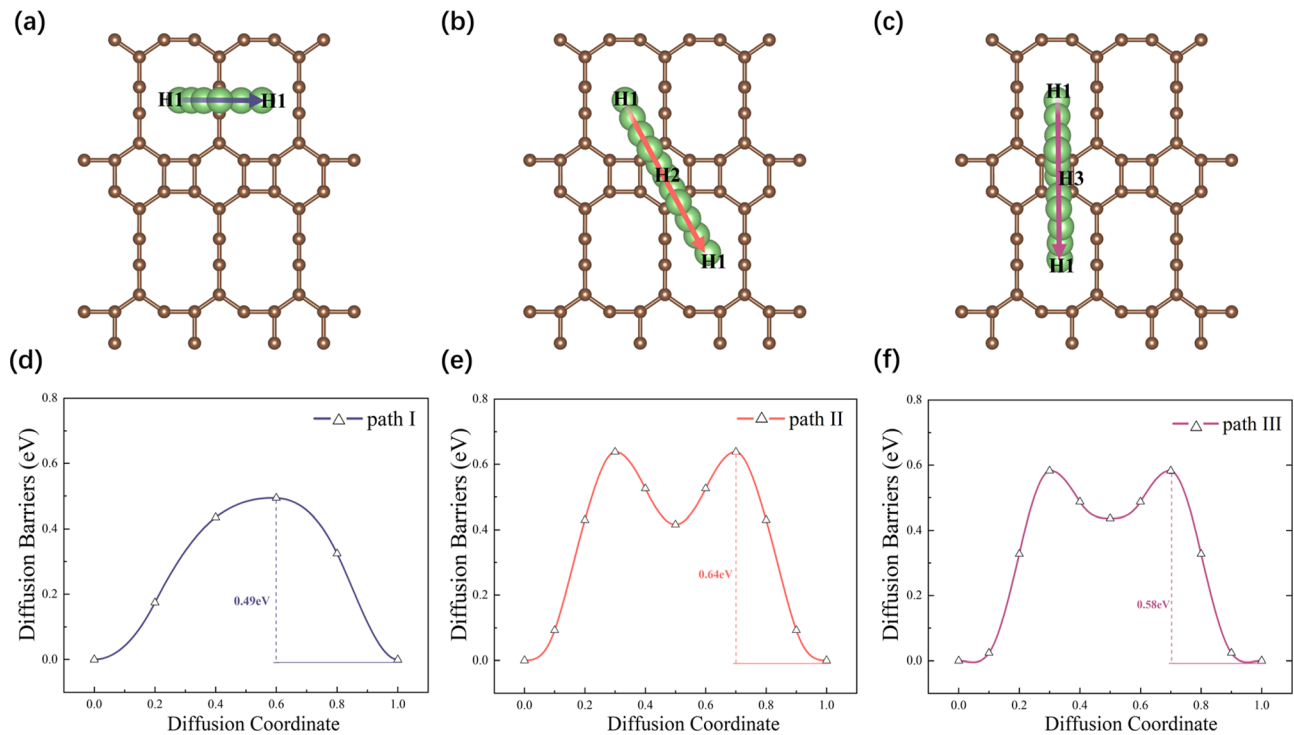


FIG. 5. (a)–(c) Three possible migration pathways on THD-C. (b)–(f) Corresponding diffusion energy profiles for Li migration along the indicated paths.

and E_{Li} is the energy of a single Li atom in bulk metal. A negative value indicates exothermic and thermodynamically favorable adsorption. Figure 6(a) illustrates that the average adsorption energy remains negative within the monolayer regime, confirming the feasibility and stability of Li adsorption on THD-C. In addition, our calculations indicate that the shortest distance between Li atoms on adjacent six-membered rings is 3.81 Å, which is larger than the Li–Li dimer bond length.⁴¹

The relative formation energies of $\text{Li}_x\text{THD-C}$ at different Li concentrations, calculated to assess thermodynamic stability, are defined as

$$E_{\text{f}} = E_{\text{Li}_x\text{THD-C}} - \frac{1}{x} [xE_{\text{Li}_{x_{\text{max}}\text{THD-C}}} + (x_{\text{max}} - x)E_{\text{THD-C}}], \quad (4)$$

where $E_{\text{Li}_x\text{THD-C}}$ and $E_{\text{Li}_{x_{\text{max}}\text{THD-C}}}$ represent the energy of THD-C with x and x_{max} Li atoms. $E_{\text{THD-C}}$ is the energy of pristine THD-C. Formation energies are analyzed via the convex hull method. Figure 6(b) indicates that the configurations at $x = 1, 2, 3, 4, 10$, and 12 lie on the convex hull and are thus thermodynamically stable, whereas the others are metastable. For the six stable phases, lattice constant variations with Li concentration are analyzed (Fig. S5), revealing a gradual expansion up to 1.51% at 12 Li coverage.

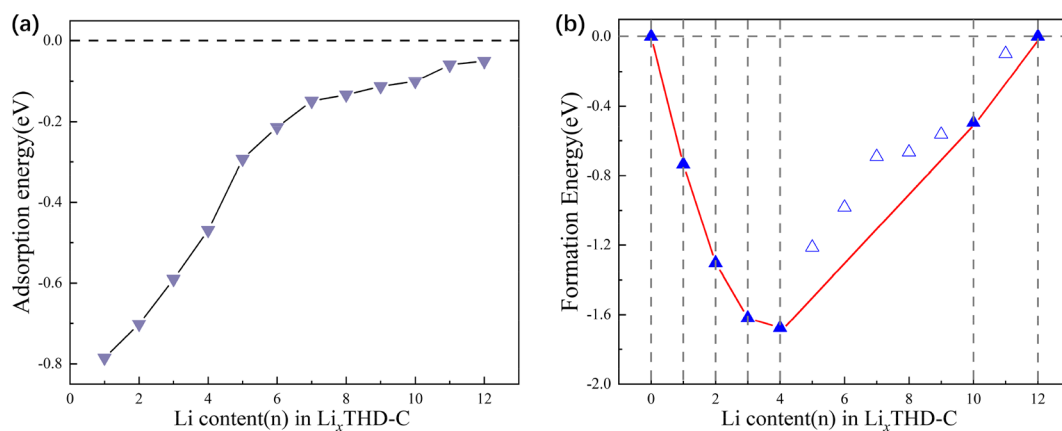


FIG. 6. (a) Average adsorption energy vs the number of lithium atoms adsorbed ($3 \times 2 \times 1$ supercell). (b) The formation energy convex hull of THD-C.

To further explore the maximum Li storage capacity of THD-C, we extend the model to include double-layer adsorption based on the 12 Li configuration. The final structure accommodates 24 Li atoms, while triple-layer adsorption fails to converge due to structural instability. The average adsorption energy remains negative (-0.11 eV). Ultimately, the nearest-neighbor Li-Li distance is 3.49 Å, larger than the Li-Li bond distance in Li dimers that reduces interlayer Li-Li electrostatic repulsion. The volume expansion for the 24 Li configuration

is only 3.23%. This value is significantly lower than that of commercial graphite anodes (12%),⁵¹ which thus guarantees favorable long-term cycling stability. The OCV is defined as

$$OCV = \frac{-E_{Li_{n_2}THD-C} + E_{Li_{n_1}THD-C} + (n_2 - n_1)E_{Li}}{(n_2 - n_1)e}, \quad (5)$$

where $E_{Li_{n_1}THD-C}$ and $E_{Li_{n_2}THD-C}$ are the total energies of the systems with n_1 and n_2 Li atoms ($n_2 > n_1$), respectively, and E_{Li} is the energy of a single Li atom in bulk. Only thermodynamically stable configurations are considered in OCV calculations (Fig. 7). The results reveal that the OCV at the 24 Li adsorption state is 0.40 V, which falls within the desirable range for anode materials (0.2–1.0 V).⁵² Compared to some commercial anodes (Table I), the lower OCV of THD-C is advantageous for achieving higher working voltage and energy density in full cells.

Furthermore, we estimate the theoretical specific capacity of THD-C based on this approach to evaluate its maximum Li loading capability. The theoretical specific capacity is calculated using the following formula:⁵³

$$C = \frac{nF}{M_{THD-C}}, \quad (6)$$

where n is the number of electrons adsorbed and reversibly released per supercell or formula unit, F is Faraday's constant (96485 C/mol⁻¹), and M_{THD-C} is the molar mass of the supercell (g/mol⁻¹). The results demonstrate that, under the maximum adsorption configuration considered, THD-C exhibits a maximum theoretical specific capacity of up to 1116.7 mAh/g, which is notably competitive compared to common anode materials. These results highlight the favorable structural stability of THD-C during lithiation and its strong potential as an LIB anode material.

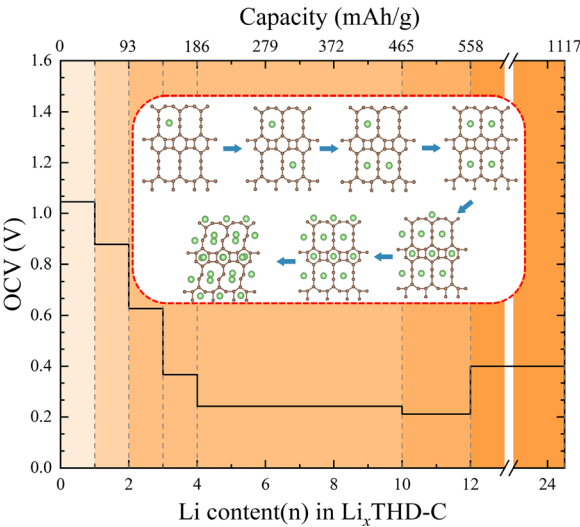


FIG. 7. OCV of THD-C at different Li adsorption concentrations. Insets show the top view of Li adsorbed on THD-C.

TABLE I. Theoretical capacity (mAh/g), diffusion barrier (eV), OCV (V), volume change (Δ) (%), and carbon rings of 2D carbon allotropes for LIBs.

Materials	Carbon rings	Diffusion barrier (eV)	Capacity (mAh/g)	OCV (V)	Δ (%)
THD-C	4-6-12	0.49–0.64	1116.7	0.40	3%
Graphene ⁵⁵	6	0.22–0.40	372	0.11	...
Graphenylene ⁵⁶	4-6-12	0.25–0.57	1116	...	...
Tolanene ¹¹	4-6-8-12	0.61–0.71	638	0.25	...
Biphenylene ⁴¹	4-6-8	0.42–0.44	624	0.34	11%
T-graphene ⁵⁷	4-8	0.37–0.47	744	0.20	1.0%
C _{358X} ⁵⁸	3-5-8-10	0.29–0.57	1116.7	0.31	...
ψ -Graphene ⁸	5-6-7	0.27–0.31	372	0.01	...
Net-W ¹⁵	4-6-8	0.40	...	0.42	1.5%
Xgraphene ⁵⁹	5-6-7	0.44–0.49	930	0.28	...
Θ -graphene ⁶⁰	5-6-8	0.45–0.48	876.7	0.30	...
Net- τ ⁶¹	4-5-6-8-10	0.30–0.36	558	0.04	13.8%
H _d -graphene ⁶²	4-5-6-7	0.21–0.32	1395.8	0.03	...
C ₅₆₇₈ ⁶³	5-6-7-8	0.29–0.44	697	0.33	...
Net-C18 ⁶⁴	5-6-8	0.32	403	0.024	...
2D-C ₅₅₈ ⁶⁵	5-8	0.42–0.49	2727.5	0.33	...
B-graphyne ⁶⁶	5-12	0.17–0.87	525	0.73	...
Holey-graphyne ⁶⁷	4-6-24	0.28	873	...	0.5%
Orth-C ₁₄ ⁶⁸	4-6-8	0.12	956	0.22	1.3%

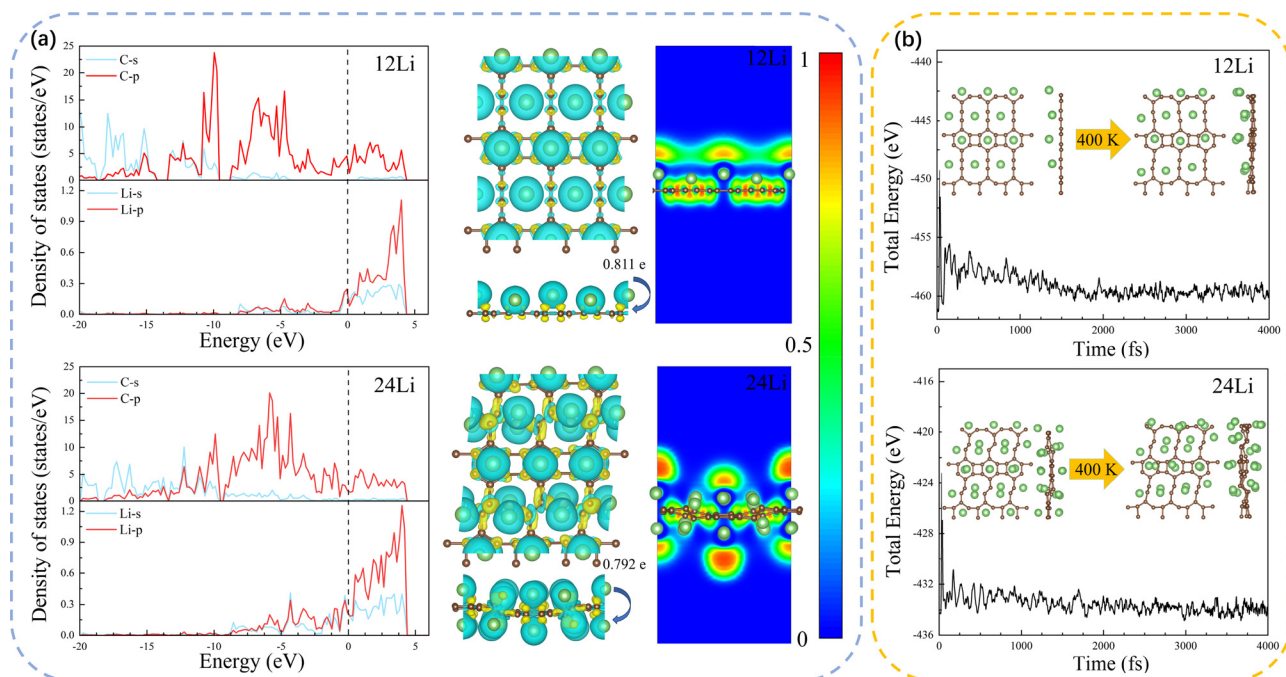


FIG. 8. (a) Electronic properties of THD-C with 12Li and 24Li adsorption, including PDOS, ELF distributions, and Bader charge transfer. (b) AIMD simulations of Li₁₂THD-C and Li₂₄THD-C at 400 K during 4 ps.

We analyze the electronic structure of THD-C with one- and two-layer Li atoms to understand the nature of the multilayer adsorption behavior. As shown in Fig. 8(a), compared with the single-Li adsorption case, the electronic states of 12 Li adsorption exhibit an apparent broadening, with this broadening becoming even more pronounced at 24 Li coverage. In addition, electron localization around Li sites weakens at 24 Li. The electrons form an extended negative electron cloud that screens the repulsive interactions between adjacent Li ions. Bader charge analysis and charge density differences reveal decreasing charge transfer per Li ($-0.910 |e| \rightarrow -0.811 |e| \rightarrow -0.792 |e|$) with increasing coverage. This trend indicates that the delocalized negative electron cloud and weakened charge transfer are crucial for stabilizing the Li-ion adsorption and facilitating the multilayer adsorption. Carbon anodes often face risks of structural instability at 80–120 °C (~ 353 – 393 K).⁵⁴ Thus, AIMD simulations are performed at 400 K for 12 Li- and 24 Li-adsorbed configurations. As depicted in Fig. 8(b), the simulations confirm that THD-C maintains its structural integrity under both single- and double-layer Li adsorption.

In summary, first-principles calculations demonstrate that THD-C possesses favorable potential as an anode material for LIBs. Our simulations indicate that $sp-sp^2$ hybridization induces p_z occupation nonuniformity, thereby enhancing Li binding in THD-C, and that two spatially adjacent acetylenic bonds form a modest Li–C coupling through the cooperative cation– π interaction via partial electron transfer, which exhibits a suitable Li adsorption energy and promotes Li-ion adsorption. Furthermore, for multilayer adsorption, the delocalized negative electron cloud and weakened charge transfer play a crucial role in stabilizing Li-ion adsorption and enabling such multilayer adsorption. THD-C exhibits a high theoretical capacity (1116.7 mAh/g),

a suitable diffusion barrier (0.49 eV), a moderate open-circuit voltage (0.40 V), and a small volume change (3.23%). In addition, AIMD simulations at 400 K confirm its outstanding thermal robustness under multilayer lithiation. These findings highlight the promise of THD-C as a high-capacity anode material for LIBs. The Li-ion storage mechanisms provide theoretical guidance for the rational design of $sp-sp^2$ hybridized two-dimensional carbon allotropes as promising anode materials.

See the [supplementary material](#) for the band structures, PDOS, the charge density difference map of the T1 site, COHP curves of Li–C interactions at H1 sites, p_z (π) electron occupancy of carbon atoms of THD-C, and AIMD simulation of THD-C involving the removal of all adsorbed Li ions from Li₂₄THD-C.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Zihao Wang: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Haoxuan Chen:** Formal

analysis (equal); Methodology (equal); Software (equal); Visualization (equal). **Zhenjie Zhao:** Formal analysis (equal); Methodology (equal); Software (equal); Visualization (equal). **Huadong Zeng:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Resources (equal); Software (equal); Supervision (equal); Visualization (equal); Writing – review & editing (equal). **Xinlu Cheng:** Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Resources (equal); Software (equal); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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